Digital Scans and Human Identification

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PRODUCT REPORT

Digital scans and human identification

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BACKGROUND

The exponential growth of digital technology in dentistry is inherently accompanied by a significant expansion of 2D and 3D dental image records. Traditional stone models are impractical to keep long-term due to storage volume and fragility.

Comprehensive and accurate models offer an excellent record of the preoperative dentition for the complete restoration of a smile that matches the original (Renne, Evans et al. 2017, Revilla-Leon, Raney et al. 2020). The longer-term storage of dental models facilitates resolving legal cases, and might aid bite mark analysis in some criminal cases (Khatri, Daniel et al. 2013). Yet another application of dental models is for human identification. In addition to DNA and fingerprints, dental examination is a primary tool for disaster victim identification (DVI) (Interpool, Tsokos, Lessig et al. 2006). Dental models that are discarded or lost may deprive biologically driven oral rehabilitation of historical tooth, bite, and bone reference points, and may hamper positive identification.

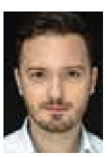
Population-wide databases for fingerprints (Peralta, Triguero et al. 2016, Thalesgroup.com 2021) and DNA (Smith 2006, Amankwaa and McCartney 2018) are limited and very fragmented. After the 2004 tsunami disaster in Thailand, 46% of the victims were identified by dental records, as compared to only 19% by DNA and 34% by fingerprints. The dental identification method is an analogous visual comparison of the ante- and post-mortem dental records (Miki, Muramatsu et al. 2017, Alsalamah and Nuzzolese 2020). The basis of this concept is that dental treatments are always very specific and unique (Pretty and Sweet 2001, Ata-Ali and Ata-Ali 2014). To confuse matters somewhat, teeth are continually impacted by abrasion, disease, trauma, and dental treatment. Thus, the available antemortem data might not correlate well to the postmortem data. Furthermore, treatment notation and information are not standardized, and it is exceedingly difficult to run an automatic search in a large, fragmented database.

Identifying the victim's dentist, or, at the very least, the area where the victim was treated, is a mandatory prerequisite for a DVI search. Antemortem dental records can be very challenging if no other victim information is available. In fact, younger patients may have only orthodontic records. The search process can be accelerated dramatically by accessing the ever-increasing number of digital scans and cloud-based data storage systems.

Digital dental records must be retained, depending on national regulations, from years to decades (Charangowda 2010, Devadiga 2014). Thus, digital dental records open new pathways for DVI. The logical next step is to find oral cavity characteristics with universality, uniqueness, invariability (stable throughout the life), and ease of access.

Monozygotic (MZ) twins cannot be distinguished by DNA analysis (Bell and Spector 2011) and they look very similar (phenotypes). Hence, one way to prove the uniqueness of an identification method is its ability to reliably distinguish MZ twins. It has recently been revealed that palatal morphology (palatal vault and surface texture) can differentiate MZ twins through intraoral scans (Simon, Liptak et al. 2020). Rugoscopy (also known as palatoscopy, calcorrugoscopy) is based on the difference in palatal rugae pattern, and can distinguish among ethnic and race groups, offering great assistance during DVI (Hermosilla Venegas, San Pedro Valenzuela et al. 2009, Bajracharya, Vaidya et al. 2013, Azab,

3 lines too long

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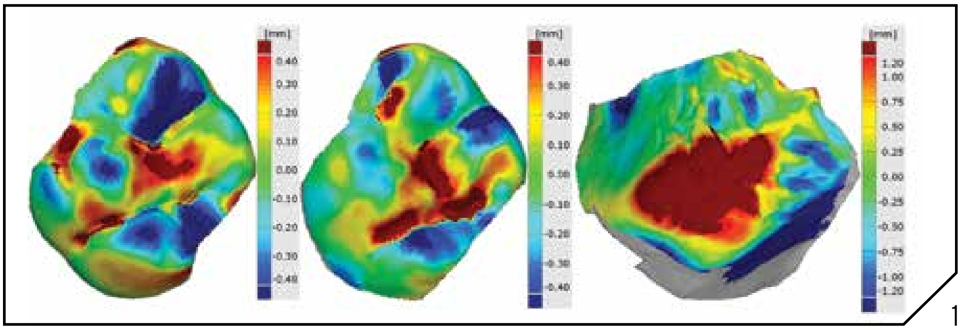

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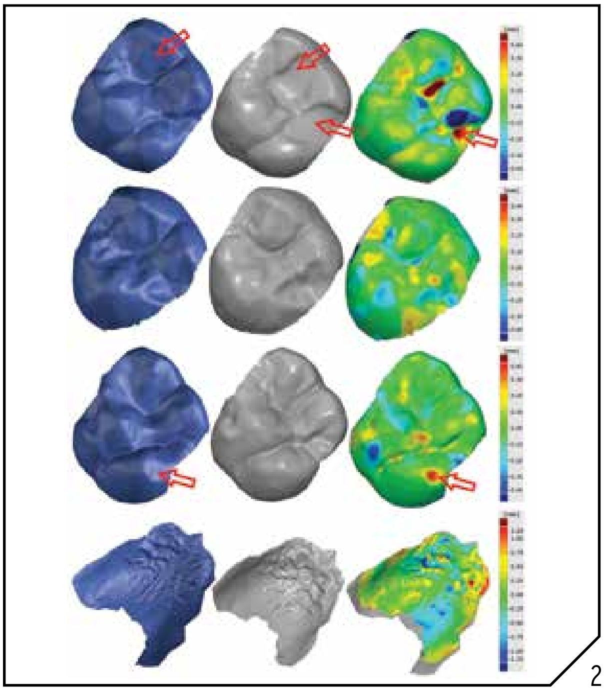

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1. The surface comparison maps of nonrelative subjects of the maxillary first molars, of the maxillary second molars, and of the palate. None of the teeth had restorations. Deep red and blue areas indicate distance deviation higher than the range of the color scale. **2.** The digital cast of the two siblings (green and blue) and the surface comparison (color map). The first row shows two maxillary first molars of the 22-year-old MZ pair (pair #3). The teeth have occlusal restorations resulting in increased deviation (red arrows). The second row shows two maxillary second molars of pair #3. These teeth have no restorations or abrasions. However, the occlusal surface morphology is quite different between siblings. The third row shows two maxillary first molars of 19-year-old MZ pairs (pair #1). There are no restorations, and the cusp shapes are very similar. However, a sign of abrasion can be seen in sibling A (blue), creating an increased deviation between scans (red arrows). The fourth row shows the palate of pair #3. Please note that the color range in the palatal is three times more than in the tooth map.

Magdy et al. 2016, Suhartono, Syafitri et al. 2016, Kommalapati, Katuri et al. 2017, Saadeh, Ghafari et al. 2017, Barbo, Azeredo et al. 2018, Basman, Puspita et al. 2018). The palate is more resistant to burn deformation injury when compared to the skin (Muthusubramanian, Limson et al. 2005). It is stable over time and little varied after orthodontic treatment (Bailey, Esmailnejad et al. 1996, Abdel-Aziz and Sabet 2001, Ali, Shaikh et al. 2016, Lanteri, Cossellu et al. 2020).

OBJECTIVE

The aim of this pilot study was to compare teeth and palate uniqueness using the intraoral scans (IOS) of MZ twins.

METHODS

Three MZ pairs, ages 17, 22, and 26 were enrolled in the study. The complete maxillary arch, including the palate, was scanned by the Emerald intraoral scanner (Planmeca, Helsinki, Finland, software version: Romexis 5.2.1). The palate was carefully isolated on each scan and was exported to a separate model. The left maxillary first molar was intact in five subjects and filled in one subject. The left maxillary second molar was intact in each subject. These two teeth were segmented, and the images were exported to respective new files.

Palatal digital models and tooth digital models were aligned between non-relatives *(Fig. 1)* and between siblings. *(Fig. 2)* The superimpositions were made using the GOM Inspect software (GOM GmbH, Germany), utilizing the local best-fit algorithm. The mean absolute deviations were calculated for each superimposition with the surface comparison tool. The data were statistically analyzed by the generalized linear mixed method using SPSS (IBM SPSS Statistics for Windows, Version 27.0., United States).

RESULTS AND DISCUSSION

The mean absolute deviations (± the standard deviation) of the first and second molars between non-relatives

(Fig. 1) were not significantly different (0.259±0.039 mm, 0.277±0.037 mm, $p=0.733$), but the mean absolute deviation of the palates was significantly higher (1.061±0.314 mm, $p<0.001$). Previous studies found that a single tooth's trueness is between 14-72µm (Lee, Jeong et al. 2017, Uhm, Kim et al. 2017, Vag, Nagy et al. 2019). The trueness of the palate was reported between 80.5µm (Zhongpeng, Tianmin et al. 2019) and 130.5µm (Gan, Xiong et al. 2016). Accordingly, the intraoral scan can distinguish between non-relative people based on either tooth or palate imagery.

Molars in MZ siblings look very similar (*Fig. 2* first three rows). The mean absolute first molar deviation between siblings was significantly lower than the second molar deviation (0.087±0.032 mm, 0.137±0.038 mm, $p<0.05$). Notwithstanding that one of the first molars of the MZ pairs had restorations, these values were significantly ($p<0.001$) lower than the deviations between non-relatives. Since these values are not much higher than the IOS trueness, they jeopardize the confidence in MZ twin identification. The palatal deviation between siblings was 3-4 ti mes higher (0.393±0.079 mm, $p<0.001$) than the teeth deviation. Although, the deviation is significantly ($p<0.001$) lower than the values between non-relatives, it was ten times higher than the precision (i.e., reproducibility, 35µm) of a recent IOS (Simon, Liptak et al. 2020). It was three times more than the trueness of the IOS regarding the palate (Gan, Xiong et al. 2016, Zhongpeng, Tianmin et al. 2019).

Along with the present findings, there is increasing evidence that the 3D digital palatal model could serve as a highly reliable tool for human identifications (Gibelli, De Angelis et al. 2018, Lanteri, Cossellu et al. 2020), and for distinguishing between MZ twins (Taneva, Evans et al. 2017, Simon, Liptak et al. 2020). Recent and ongoing developments in IOS technology will further improve the reliability. Digital casts of twins can also be used to study genetic and environmental factors in odontogenesis (Smith, Townsend et al. 2009). It is of great importance that dentists should not discard digital models after completion of the dental work. These archived models are very useful for legal, forensic, and rehabilitation purposes, now and far into the future.

Oral Health welcomes this original article.

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